



Multimodal Quantum-inspired Network for Emotion Recognition

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ABSTRACT

Emotion Recognition in Conversations (ERC) is a prominent research topic in Natural Language Processing (NLP). Most current approaches rely on deep black-box models, which lack sufficient interpretability. To this end, some work inspired by quantum theory has been proposed, trying to explore new solutions with the help of uncertainty modeling. However, these methods fail to fully integrate useful information within conversations, and overlook many effective classical mechanisms when employing quantum probabilistic modeling. In this paper, we propose a multimodal quantum-inspired network (MQN) to tackle these challenges from the perspectives of feature embedding and information fusion. We design a conversation-oriented feature embedding (CFE) method to effectively encode useful information in conversations and establish global interaction through phase embedding. Additionally, we introduce a joint information fusion (JIF) method that incorporates attention and gating mechanisms to enhance intra-modal feature representation and inter-modal interactions in multimodal fusion. Experimental results on two benchmark ERC datasets demonstrate that our model outperforms other quantum-inspired models and achieves performance comparable to state-of-the-art methods, providing a novel perspective for conversation emotion recognition.

1. Introduction

Emotion Recognition in Conversations (ERC) is a research hotspot in the field of artificial intelligence, which uses Natural Language Processing (NLP) methods to analyze the emotional information contained in the conversation process. With the popularization of the Internet, the rapid development of the short video industry and instant messaging have provided abundant materials for emotion analysis, attracting widespread attention to emotion recognition (Chen et al., 2018; Hazmoune & Bougamouza, 2024; Li et al., 2024a,b; Song et al., 2023; Yeh et al., 2019; Zhou et al., 2018). Fig. 1 is a flowchart example for conversational emotion recognition. In addition to text, conversations also contain useful information such as facial expressions (Ekman, 1993) and speaking intonation, providing different perspectives for emotion analysis. These pieces of information are fed into a neural network for the extraction of multimodal features, which are then used to determine the emotions contained in the conversation. Current research typically employs methods such as memory networks (Hazarika et al., 2018a,b; Shen et al., 2021) and attention mechanisms (Chauhan et al., 2019; Ji et al., 2022; Jiao et al., 2020; Zhang et al., 2022b) to model contextual information, speaker information, and emotional data within conversations,

thereby integrating multimodal features to enhance the performance of emotion computation.

The challenge faced by these popular deep neural networks is their lack of sufficient interpretability due to their black-box structure (Baltrušaitis et al., 2018). The introduction of quantum theory provides new and effective approaches for language understanding tasks. Studies have shown that the underlying mathematics of quantum theory provides a better explanation of human cognition compared to traditional models (Busemeyer & Bruza, 2012). Since conversational emotion analysis is fundamentally a human cognitive problem, leveraging quantum theory to enhance models could potentially offer more advantages in emotion recognition (Li & Hou, 2021; Liu et al., 2023). In the field of conversational emotion recognition, many scholars have been inspired by quantum entanglement (Gkoumas et al., 2021b), quantum interference, or quantum measurement (Zhang et al., 2020) to develop models for improved multimodal representation learning. These quantum-inspired models represent utterances in the form of quantum states, which to some extent enhance the interactions between different utterances.

Despite the advantages over traditional models, existing quantum-inspired models still face several challenges in the task of multimodal

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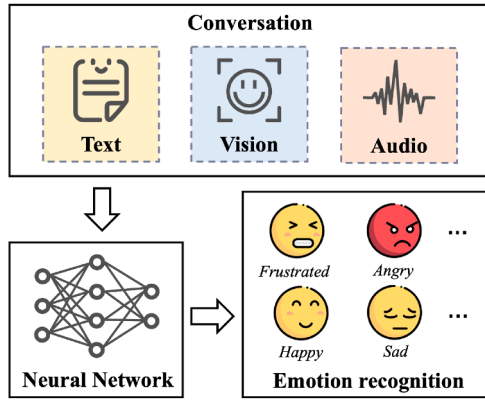


Fig. 1. The flowchart example for conversational emotion recognition.

conversational emotion recognition. Firstly, few studies ensure the comprehensive integration of useful information within the conversation. Specifically, in multimodal feature extraction, the modeling of utterance position information is often overlooked (Li et al., 2023; Zhang et al., 2019). Moreover, many studies fail to consider all relevant modal types (Zhang et al., 2020). Secondly, many studies focus excessively on quantum-like representations, which has led to the underutilization of some classical effective mechanisms. This may weaken the model's ability to represent features within each modality and the interaction between modalities during multimodal fusion.

In this paper, we construct a multimodal quantum-inspired network (MQN) to address these challenges. First, in order to integrate useful information from the conversation as fully as possible, we propose a conversation-oriented feature embedding (CFE) method. In addition to multimodal information and speaker information, we also consider utterance position information. These useful data are efficiently encoded into quantum states through amplitude embedding and phase embedding, establishing a global interaction between multimodal features and contextual cues. Second, we propose a joint information fusion (JIF) method that leverages effective classical mechanisms from both intra-modal and inter-modal perspectives to enhance information extraction during the multimodal fusion process. Specifically, intra-modal information is effectively extracted by introducing attention mechanisms, while for inter-modal information extraction, we integrate a gating mechanism into a quantum-like recurrent neural network (QRNN) (Li et al., 2021b) to construct a quantum-like gated recurrent unit (QGRU), which models the evolution of complex dependencies in the conversation. At the output of the MQN, the evolved density matrix undergoes quantum-like measurements, collapsing into the corresponding basis state of each sentiment label, thereby enabling the recognition of the emotional category of each utterance.

The main contributions can be summarized as follows:

- We propose the CFE method that effectively integrates useful information from the conversation. By employing amplitude and phase embeddings, this method encodes the information into quantum states, establishing global interactions.
- We design the JIF method to enhance information extraction during the multimodal fusion process. Specifically, we incorporate the attention mechanism to extract intra-modal information and introduce the gating mechanism to construct a quantum-like GRU for modeling the evolution of discourse emotion.
- Based on CFE and JIF, we construct a multimodal quantum-inspired network called MQN for conversational emotion recognition. The experimental results on the MELD and IEMOCAP datasets show that MQN outperforms other quantum-inspired models and is comparable to state-of-the-art methods in conversational emotion recognition.

The remaining content of this paper can be summarized as follows. In Section 2, foundational knowledge of quantum computing and pertinent methods for conversational emotion recognition are introduced. Section 3 offers a comprehensive overview of the proposed model MQN. The model performance is validated with conversational Emotion Recognition experiments, and the results are analyzed in Section 4. The conclusions and potential avenues for future research are drawn in Section 5.

2. Related Works

2.1. Preliminaries

In this paper, we employ the framework of quantum mechanics for research. Subsequently, fundamental concepts of quantum mechanics will be presented, which include *quantum state* and *quantum measurement*.

2.1.1. Quantum state

Quantum state refers to the state of a quantum system, containing all the relevant information of the system. Quantum states can be classified into *pure state* and *mixed state*.

When an isolated system is composed of only one particle, its state can often be described as a pure state, which can be represented as

$$|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (1)$$

where " $|\cdot\rangle$ " represents the *Dirac notation*, which is the standard notation used in quantum mechanics to describe states. α and β are complex numbers, subject to the condition $|\alpha|^2 + |\beta|^2 = 1$. When $|\varphi\rangle$ is measured, the probability of obtaining $|0\rangle$ is $|\alpha|^2$, while the probability of obtaining $|1\rangle$ is $|\beta|^2$.

Mixed state is a probabilistic mixture of a series of pure states, which can be represented by the *density matrix*:

$$\rho = \sum_{i=1}^n p_i |\varphi_i\rangle \langle \varphi_i|, \quad (2)$$

where $\{|\varphi_i\rangle\}_{i=1}^n$ is the set of all possible quantum states, p_i corresponds to the probability of each quantum state $|\varphi_i\rangle$, and satisfies $p_i \geq 0$, $\sum_{i=1}^n p_i = 1$. If the density matrix of a system can be written as $\rho = |\varphi\rangle \langle \varphi|$, then the quantum state of the system is a pure state.

2.1.2. Quantum measurement

Quantum measurement can cause the collapse of quantum states, altering the state of the quantum subsystem. Multiple measurements of the same quantum system may produce different results. Since the process of quantum measurement is irreversible, it is impossible to deduce the initial state of the system before measurement based on the measurement results.

Quantum measurement can observe quantum systems in a mixed state ρ . During the measurement process, an observable O is required, which is composed of a set of measurement operators $\Pi_i = |m_i\rangle \langle m_i|$. $|m_i\rangle$ forms a complete orthonormal basis in the Hilbert space and also serves as the eigenvectors of O . According to Born's rule, the probability of obtaining the value m_i after measurement is as follows:

$$p_i = \text{Tr}(\rho |m_i\rangle \langle m_i|) = \langle m_i | \rho | m_i \rangle, \quad (3)$$

the measurement results satisfy the formula $\sum p_i = 1$.

2.2. Complex-valued Word Embedding

In the field of NLP, word embedding is a method that can represent words or utterances in the form of vectors. In classical scenarios, the values in the vectors are real numbers. Common word embedding methods include one-hot encoding, Word2Vec (Mikolov et al., 2013), neural network language models (Bengio et al., 2003), and so on. In quantum scenarios, there is a corresponding concept, where complex-valued word

embeddings can represent words as quantum states in Hilbert space, expressed as:

$$|\varphi\rangle = \sum_{j=1}^n z_j |k_j\rangle = \sum_{j=1}^n r_j e^{i\theta_j} |k_j\rangle, \quad (4)$$

where $\{z_j\}_{j=1}^n$ are complex-valued vectors, $\{r_j\}_{j=1}^n$ are the amplitude vectors of $\{z_j\}_{j=1}^n$ (satisfying $\sum_{j=1}^n r_j^2 = 1$), $\{\theta_j\}_{j=1}^n$ are the phase vectors (satisfying $\theta_j \in [-\pi, \pi]$), and $\{|k_j\rangle\}_{j=1}^n$ are the orthonormal basis set within the Hilbert space.

The amplitude vectors $\{r_j\}_{j=1}^n$ and the phase vectors $\{\theta_j\}_{j=1}^n$ can represent different meanings in the complex-valued word embedding. The amplitude of the complex value can be utilized to express the semantic details within the word, while the phase can have multiple meanings, indicating emotional information (Zhang et al., 2018b) or polysemous word information (Wang et al., 2019a).

Complex-valued word embedding is often used in information retrieval (Zuccan et al., 2011), emotion recognition (Li et al., 2021a; Liu et al., 2021; Zhang et al., 2018b), link prediction (Melucci et al., 2015; Trouillon et al., 2016), and question-answering (Li et al., 2019) fields. Wang et al. (Wang et al., 2019a) utilized complex-valued vectors to describe the semantics of compound words. Based on this, they proposed an end-to-end neural network for text classification. Zuccan et al. (Zuccan et al., 2011) found that complex numbers have advantages in describing complex information, which can make information retrieval more effective. Zhang et al. (Zhang et al., 2018b) employed complex-valued word embedding to describe data from different channels in multimodal sentiment analysis and introduced a quantum-inspired multimodal sentiment analysis (QMSA) method. Chu et al. (Chu et al., 2024) encode more semantic information using complex-valued word embedding techniques to better determine sentiment in conversations. Li et al. (Li et al., 2019) constructed a network with complex-valued elements for semantic matching.

However, the current methods for complex-valued word embedding used in conversational emotion recognition tasks neglect the modeling of utterance position, which may result in insufficient integration of effective information.

2.3. Quantum-inspired Neural Network

Up till now, many scholars have been inspired by concepts in quantum computing, introducing ideas such as quantum gates (Li & Zhao, 2018), tensor networks (Patel et al., 2022), and complex-valued weights (Shi et al., 2022) into neural networks to improve performance. As a result, a series of quantum-inspired neural networks have been proposed. To address challenges present in classical computing, quantum-inspired neural networks are widely applied in information retrieval (IR) (Jiang et al., 2020; Upreti et al., 2020), question-answering (Chen et al., 2021; Li et al., 2019; Zhang et al., 2025a,b, 2018a), emotion recognition (Li & Hou, 2021; Liu et al., 2023; Shi et al., 2021, 2024), Medical diagnosis (Swathi et al., 2024), and other fields. For example, Zhang et al. (Zhang et al., 2018a) employed density matrices for word embedding representation, introducing a more effective neural network based quantum-like language model (NNQLM) for question-answering matching. Building upon this work, Shi et al. (Shi et al., 2021) introduced an interpretable end-to-end quantum-inspired neural network to extract language features from the text, achieving promising results in text emotion recognition tasks. To accelerate the diagnostic process of thyroid cancer, Swathi et al. (Swathi et al., 2024) employed a quantum-inspired convolutional neural network to improve the classification accuracy of thyroid nodules.

In conversational emotion recognition, some quantum-inspired neural networks have also emerged. Gkoumas et al. (Gkoumas et al., 2021a) introduced a novel quantum cognitively motivated fusion strategy, describing utterances using positive and negative quantum superposition states and modeling single-modal classifiers as mutually incompatible observables. This work is the first to explore and model the quantum

cognitively motivated approach to emotional judgment incompatibility in video emotion analysis. Furthermore, Gkoumas (Gkoumas et al., 2021b) also discovered that quantum entanglement can be used to describe the inseparability between modalities, leading to the proposal of an entanglement-driven fusion neural network. Experimental results show that the model achieves significant performance improvement. Li et al. (Li et al., 2021b) compared the process of emotion recognition with quantum measurement, constructing a complex-valued neural network to achieve conversational emotion recognition, thereby improving training accuracy and efficiency. Although these quantum-inspired neural networks for conversational emotion recognition have made excellent progress, they pursue as complete quantum-like representation as possible, but ignore some classical mechanisms that have been proved to be effective.

3. Methodology

This section introduces a multimodal quantum-inspired network named MQN for conversational emotion recognition. MQN integrates quantum theory into natural language processing, aiming to better extract useful information during the conversation. Fig. 2 illustrates the overall framework of MQN.

First, the CFE method is used to fully extract useful information from the conversation. Specifically, CFE encodes multimodal information through amplitude embedding, and uses phase embedding to encode speaker information and utterance position, thereby efficiently representing the information contained in each utterance in the form of a quantum state. Subsequently, the JIF method is used to mix the unimodal pure states obtained from CFE, where the Transformer encoder is introduced to extract intra-modal information, and the QGRU performs sufficient inter-modal information extraction in a quantum evolution manner. Finally, the evolved mixed state is measured to determine the emotional category of the utterance. For ease of problem description, we provide annotations for commonly used symbols, as shown in Table 1.

3.1. Conversation-oriented Feature Embedding

We believe that the useful information in the dialogue process includes multimodal information, speaker information, and utterance position information. Multimodal information refers to the emotional information contained in the three modalities of text, vision and audio, with features of each modality being extracted individually. Considering the impact of personality and speaking habits, the participants in the conversation also encompass valuable information (Ghosal et al., 2019; Majumder et al., 2019; Zhao et al., 2018). At present, some scholars have proved the importance of position information in conversational emotion recognition through experiments (Wang et al., 2019b; Zhang et al., 2022a). However, in most of the previous studies, the conversational position information is often ignored, and there is a lack of methods to make full use of these three types of information in the conversation. Therefore, we propose the CFE method to solve this problem.

CFE is a complex-valued embedding method proposed specifically for conversations, designed to encode multimodal information, speaker information, and utterance position. According to the concept of complex-valued word embedding described in Section 2, the

Table 1
Definition of Symbols in Conversation.

Notation	Description
n	The number of utterances in the conversation
u_j	The j th utterance in the conversation, $j \in \{1, 2, \dots, n\}$
a_j, v_j, t_j	Acoustic, visual and textual feature of the j th utterance
pos	The position of each utterance in the conversation
spe	The speaker of each utterance

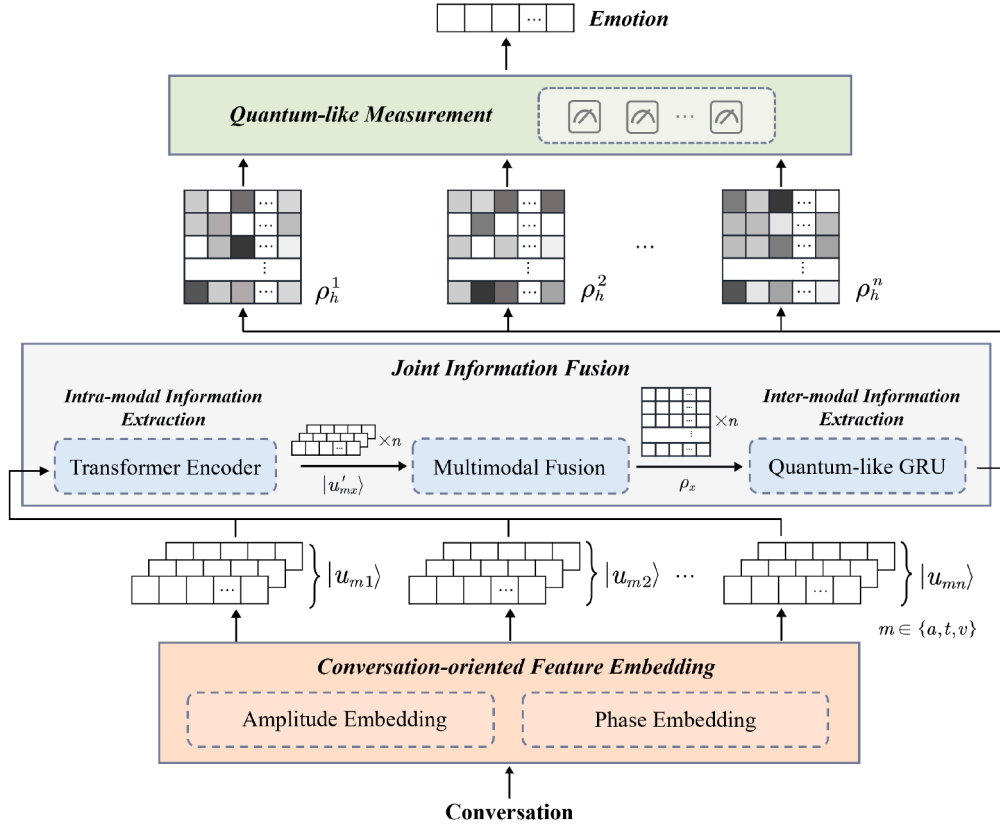


Fig. 2. Architecture of MQN model. The CFE method encodes useful information in the conversation through amplitude embedding and phase embedding. Then, the JIF method integrates information from three modalities while effectively extracting both intra-modal and inter-modal information. Finally, quantum-like measurements are used to determine the emotional category of the utterance.

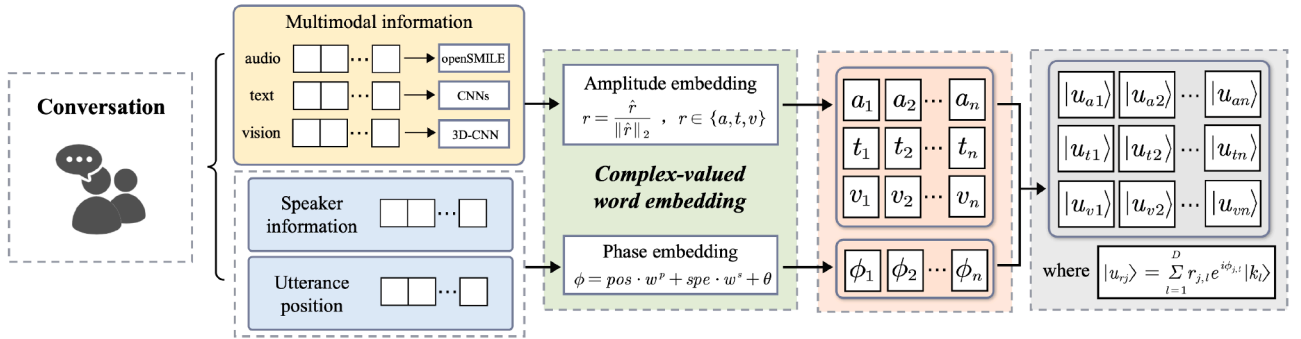


Fig. 3. Architecture of the conversation-oriented feature embedding method.

amplitude and phase of complex numbers can naturally encode information such as word meaning and word order, while also addressing the issue of polysemy. Therefore, we encode conversation information into the Hilbert space through amplitude embedding and phase embedding. The architecture of the CFE method is shown in Fig. 3.

Amplitude embedding is used to encode information from text, vision, and audio. The pre-extracted features are obtained through different neural networks after the multimodal information input. Following a similar setup as Hazarika *et al.* (Hazarika *et al.*, 2018b), we use convolutional neural networks (CNNs) to extract text features, openSMILE to extract audio features, and a 3D-CNN to extract visual features. This results in multimodal features $\{\hat{r}_j\}_{j=1}^n$ ($\hat{r}_j \in \{\hat{a}_j, \hat{t}_j, \hat{v}_j\}$, where $\hat{a}_j, \hat{t}_j, \hat{v}_j$ represent acoustic, textual, and visual features, respectively). By inputting

$\hat{r}_j$ into the amplitude embedding, the normalized output $r_j = \frac{\hat{r}_j}{\|\hat{r}_j\|_2}$ can be obtained, where $r_j \in \{a_j, t_j, v_j\}$.

Phase embedding can encode both speaker information and utterance position into the phase vector, represented as $\phi_j = pos \cdot w_j^p + spe \cdot w_j^s + \theta_j$, where w_j^p is the frequency of utterance position, w_j^s is the frequency of specific speaker, and θ_j is the initial phase with all elements $\in [-\pi, \pi]$. The phase vector $\{\phi_j\}_{j=1}^n$ is jointly influenced by pos , spe , and θ_j . The frequencies w_j^p and w_j^s represent the degree of influence of utterance position and speaker information respectively and can be continuously updated during the training process. The higher the value of w_j^p , the greater the impact of changes in utterance position on the embedding, i.e., the more distinct the feature differences between different

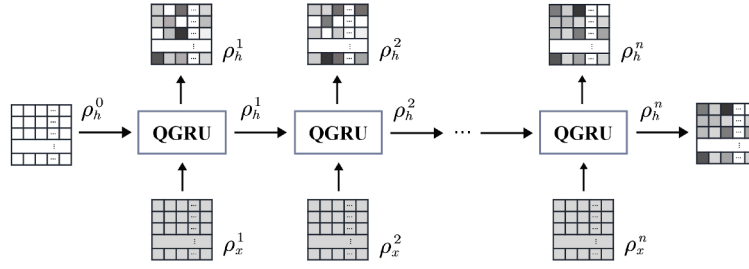


Fig. 4. Architecture of quantum-like GRU.

utterances. A higher value of w_j^s indicates a greater influence of the speaker on the conversation.

In this way, each modality of each utterance can be encoded as a pure state, defined as

$$|u_{aj}\rangle = \sum_{l=1}^D a_{j,l} e^{i\phi_{j,l}} |k_l\rangle, \quad (5)$$

$$|u_{tj}\rangle = \sum_{l=1}^D t_{j,l} e^{i\phi_{j,l}} |k_l\rangle, \quad (6)$$

$$|u_{vj}\rangle = \sum_{l=1}^D v_{j,l} e^{i\phi_{j,l}} |k_l\rangle, \quad (7)$$

where $\{|k_l\rangle\}_{l=1}^D$ is an orthonormal basis of the Hilbert space, and D is the dimensionality of the complex-valued embeddings. By encoding speaker information and utterance position in the phase, complex nonlinear interactions between multimodal features and contextual cues can be generated, thereby giving the model a stronger understanding of the contextual relationship and better capturing speaker-related features.

3.2. Joint Information Fusion

The JIF method considers both intra-modal and inter-modal aspects. It utilizes the Transformer encoder to effectively extract internal features of each modality before multimodal fusion and employs the QGRU to extract inter-modal features after fusion.

3.2.1. Intra-modal Information Extraction and Fusion

Considering that the contribution of each modality varies across different utterances, i.e., the modality responsible for determining the sentiment category may differ by utterance, we first consider each modality separately. To more effectively model the significance of each modality within an utterance, we employ a Transformer encoder (Vaswani et al., 2017) with an effective mechanism of attention to extract intra-modal information. Concretely, for each modality $m \in \{a, t, v\}$, the complex sequence is represented as $|u_{mj}\rangle \in \mathbb{C}^D$ (from CFE in Section 3.1). We apply the encoder to the real and imaginary parts independently, then recombine them to obtain the refined unimodal pure states $|u'_{aj}\rangle$, $|u'_{tj}\rangle$, and $|u'_{vj}\rangle$. The encoder comprises two identical layers, where each layer consists of a self-attention mechanism and a position-wise feed-forward network, and its parameters are shared across modalities and across real/imaginary components to maintain parameter efficiency. This design not only helps capture contextual and semantic information for each modality, but also keeps the computational and memory overhead modest at the data scale.

Before inter-modal information extraction, it is necessary to perform multimodal fusion. Multimodal fusion can combine information such as textual, visual, and acoustic features for subsequent analysis. Here, we achieve multimodal fusion by mixing each unimodal pure state. Specifically, for each utterance j , we first form unimodal density matrices by outer products $\rho_{mj} = |u'_{mj}\rangle\langle u'_{mj}|$, and then compute a convex combination with learnable weights p_{mj} to obtain the density matrix of the

mixed state, represented as follows:

$$\rho_j = p_{aj} |u'_{aj}\rangle\langle u'_{aj}| + p_{tj} |u'_{tj}\rangle\langle u'_{tj}| + p_{vj} |u'_{vj}\rangle\langle u'_{vj}|, \quad (8)$$

where p_{aj} , p_{tj} , and p_{vj} are non-negative, and $\sum_m p_{mj} = 1$. By updating the corresponding mixing weights for different utterances, the model learns the contribution of each modality to the final sentiment. Since the fusion is realized as a convex combination of unimodal density matrices, when a modality is missing or unreliable (e.g., occluded faces or noisy audio), the per-utterance mixture weights $\{p_{mj}\}$ adaptively down-weight that modality and place more emphasis on the informative ones, which mitigates sensitivity to modality availability and alignment. Note that the above multimodal fusion yields a classical mixture without cross-modality coherence terms, and therefore inter-modal interactions are modeled dynamically in the next section.

3.2.2. Inter-modal Information Extraction

To better capture the long-term dependencies in conversations, we introduce a gating mechanism based on the QRNN to construct the QGRU for tracking the evolving dynamics of emotional states in conversations. Fig. 4 illustrates the architecture of QGRU. Intuitively, the input to the QGRU is a quantum state sequence $\rho_x^1, \dots, \rho_x^N$ formed by the multimodal fused features, where N is the sequence length. For the input at the current time step ρ_x^t , its output ρ_h^t is determined by the previous hidden state ρ_h^{t-1} , the update gate z^t , the reset gate r^t , and the candidate hidden state $\tilde{\rho}_h^t$, as shown in the following expression:

$$z^t = \sigma(U_{zh}\rho_h^{t-1}U_{zh}^* + U_{zx}\rho_x^tU_{zx}^*), \quad (9)$$

$$r^t = \sigma(U_{rh}\rho_h^{t-1}U_{rh}^* + U_{rx}\rho_x^tU_{rx}^*), \quad (10)$$

$$\tilde{\rho}_h^t = \tanh(r^t U_h \rho_h^{t-1} U_h^* + U_x \rho_x^t U_x^*), \quad (11)$$

$$\rho_h^t = (1 - z^t) \rho_h^{t-1} + z^t \tilde{\rho}_h^t. \quad (12)$$

Where σ is the sigmoid activation function and $\tanh$ is the hyperbolic tangent activation function. $U\rho U^*$ denotes the quantum unitary transformation of the density matrix ρ . U_{zh} , U_{zx} , U_{rh} , U_{rx} , U_h , U_x are trainable unitary matrices. In order to ensure that the unitary property is not destroyed during the end-to-end training process, we use the Riemannian approach (Wisdom et al., 2016) to update the unitary matrix.

Compared to the QRNN used in previous methods (Li et al., 2021b), the QGRU introduces a gating mechanism (z^t and r^t) to better control the information propagation and memory between quantum states. Specifically, the z^t determines whether the information input to the hidden state needs to be retained. When its value is higher, the model tends to ignore the previously saved hidden state and focus on the new input. As the value is lower, the model relies more on the previously stored information, which helps prevent the loss of long-term memory and preserves important historical information. The r^t controls the combination of new information with previous memories. In contrast, when the r^t is smaller, more memory is ignored, allowing the model to selectively forget unimportant information and ensuring flexible handling of short-term dependencies. This means that the QGRU has more degrees of freedom to model complex dependencies, and the reduction of recursive dependencies also helps mitigate gradient issues during training, thereby lowering the difficulty of training.

Unlike GRU (Cho et al., 2014), QGRU is carefully designed to fit into a quantum-inspired learning framework. The input and output of QGRU are complex-valued density matrices. We believe that these density matrices representing probabilistic mixed states describe the classical distributions of different states in Hilbert space, thereby introducing more dynamic characteristics in the update of hidden states and enhancing the model's ability to handle nonlinear and complex relationships. The previously encoded discourse position information in the complex phase continuously participates in the global interaction through evolution, thus introducing nonlocality into the hidden state, which may enable QGRU to better capture long-distance dependencies. In addition, the reversibility of unitary transformations results in less information loss during training in the QGRU compared to the GRU, which implies lower energy consumption and stronger memory capacity.

3.3. Quantum-like Measurement

After the evolution, we use a quantum-like measurement to collapse the mixed state, which contains multiple independent emotions (i.e., pure states), in order to obtain the sentiment label for each utterance. Rather than relying on loosely specified measurement formulations, we employ an operator-based, parameterized measurement that maintains training stability while providing a unified formulation closer to quantum semantics. Here, we choose the Pauli operator σ in quantum computing for implementation.

As each measurement yields only a single value, it is necessary to determine the number of measurement operators based on the number of categories in the dataset, establishing a one-to-one correspondence between them. Different measurement operators can be constructed by applying the Pauli operator σ to different quantum bits. For instance, the measurement operator constructed by applying the σ to the first quantum bit can be expressed as $\sigma \otimes I \cdots I$ (where $\otimes$ represents the tensor product). All measurement operators collectively form an observable, denoted as O . The measurement results can be obtained by multiplying the measurement operators with the density matrix and computing the trace, which can be represented as:

$$p_i = \text{Tr}(\rho_h^i \cdot O_i), \quad (13)$$

Where O_i represents the i th measurement operator (e.g. $O_1 = \sigma \otimes I \cdots I$, $O_2 = I \otimes \sigma \otimes I \cdots I$ and so on).

At this point, the measurement results are already able to reflect the relative credibility of different sentiment labels. Finally, these preliminary measurement results are input into a Softmax layer for further normalization, converting them into a probability distribution of sentiment labels.

4. Experiments

MQN is used to conduct conversational emotion recognition tasks to verify the performance of the model. In this section, we first provide a detailed introduction to the basic settings of the experiment, then analyze the experimental results, and finally validate the effectiveness of the proposed method with the ablation study.

4.1. Experimental Settings

The experimental settings are elaborated from the aspects of datasets, evaluation metrics, baseline models, and hyper-parameters, which can be introduced separately.

4.1.1. Datasets

To evaluate the proposed model, we conducted training on two datasets for conversational emotion recognition, namely MELD (Poria et al., 2019) and IEMOCAP (Busso et al., 2008). MELD is extended and enhanced by the EmotionLines dataset, comprising three modalities: text, audio, and vision. It encompasses over 1400 dialogues and 13,000

Table 2

Distribution of MELD and IEMOCAP.

Dataset	# Conversations			# Utterances		
	train	valid	test	train	valid	test
IEMOCAP	96	24	31	6808	1702	1623
MELD	1039	114	280	9989	1109	2610

utterances from the *Friends* TV series, involving multiple participants in the dialogue process. Meanwhile, MELD is a seven-class dataset that includes seven emotional categories: joy, sad, neutral, surprise, angry, disgust, and fear. Each utterance is labeled with the corresponding emotion label. IEMOCAP primarily consists of twelve hours of improvised performance videos from two American actors, capturing and recording live speech, facial expressions, and text. It is a multi-class dataset containing information from three modalities. Annotators have labeled the emotional tags of IEMOCAP, categorizing them into six classes: happy, sad, angry, frustrated, excited, and neutral. For better training, these two datasets can be further divided into train, valid, and test sets. The number of conversations and utterances contained in each dataset is shown in Table 2.

4.1.2. Evaluation Metrics

Since a single evaluation metric may not effectively describe the model's performance, it is necessary to select multiple metrics for a comprehensive assessment. We choose *accuracy*, *precision*, and *F1 score* as the evaluation criteria for the model. Accuracy refers to the ratio of correctly predicted samples to all samples. It is an intuitive measure typically used to assess the overall performance of the model. Higher accuracy indicates better model performance. To better define accuracy, four metrics from the confusion matrix are introduced: TP, FP, FN, and TN. Therefore, the accuracy can be defined as

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}. \quad (14)$$

Precision is the ratio of correctly predicted positive samples among those predicted as positive and can be expressed as

$$Precision = \frac{TP}{TP + FP}. \quad (15)$$

The F1 score is effective in focusing on minority samples, making it suitable for assessing imbalanced datasets. It is primarily described by two metrics: precision and recall. Recall represents the ratio of successfully predicted positive samples among the actual positive samples, i.e., $Recall = \frac{TP}{TP + FN}$. Consequently, the F1 score can be expressed as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \quad (16)$$

4.1.3. Baseline Models

We selected several sentiment analysis models for comparison to better analyze the performance of the proposed model. The description of each model is provided below.

BC-LSTM (Poria et al., 2017): In order to better model contextual information in the conversation, the model utilizes Bi-directional Contextual LSTM (BC-LSTM) to extract features. A 5-10% performance improvement has been achieved in utterance-level sentiment analysis tasks.

MFN (Zadeh et al., 2018): The Memory Fusion Network (MFN) can be applied to multi-view sequential learning. In order to fully extract information from each view, this method utilizes LSTM to learn the interaction of specific views and then achieves cross-view interaction through the Delta-memory Attention Network (DMAN).

MuT (Tsai et al., 2019): The model is capable of addressing the issue of varying sampling rates across different modalities in time series analysis. The authors introduce a Multimodal Transformer (MuT) to achieve interaction between modalities in different time steps, while experimental results demonstrate the model's advancement.

ICON (Hazarika et al., 2018a): The Interactive Conversational Memory Network (ICoN) is a multimodal feature extraction framework proposed by Hazarika et al., suitable for classification and regression tasks. This method is capable of hierarchically modeling the emotions of both parties in a conversation, and then making predictions through memory networks.

DialogueRNN (Majumder et al., 2019): The model is proposed as an improvement on the basis of recurrent neural network (RNN), analyzing the emotions contained in the conversation based on the speaker and context. Moreover, the model is easily scalable and can be used as a pre-trained model for dialogue generation.

QMNN (Li et al., 2021b): This is a quantum-inspired neural network model. The authors draw an analogy between the process of sentiment analysis and quantum measurement, utilizing quantum advantages to enhance model performance. Additionally, a Quantum-like Recurrent Neural Network (QRNN) layer is proposed to extract multimodal information.

MVN (Ma et al., 2022): Multi-view networks (MVN) can perform real-time sentiment recognition in conversations from both word-level and utterance-level perspectives. This approach captures both the utterance-level dependencies within the context and the word-level dependencies between them.

4.1.4. Hyper-parameters

In order to enhance the model's performance and address issues like overfitting, it is necessary to configure hyper-parameters. We primarily optimize the results by adjusting the *batch size*, *epoch*, *learning rate*, and *dropout rate*. Batch size specifies the number of samples used in each iteration during training. Due to the disparate sizes of the two datasets, different values are chosen for batch size. For MELD, batch size options include 16, 24, and 32. In the case of IEMOCAP, batch size options are 4, 8, and 16. Epoch refers to one complete iteration over the entire training dataset. During the training process, multiple epochs can be set to learn and update parameters continuously. We found that setting the epoch to 20 yielded the best results. Learning Rate is a crucial parameter in deep learning, influencing the convergence speed of model training. In our experiments, learning rate options include 0.0007, 0.0008, 0.0009, 0.001, 0.002, 0.004, and 0.008. Additionally, the Adam optimizer is employed to adjust the learning rate. Dropout is a regularization technique that randomly deactivates a portion of neurons during training, reducing the risk of overfitting and improving the model's generalization capability. Dropout rate options include 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, and 0.8.

We construct MQN using the deep learning framework PyTorch. The experiments were carried out on a Linux server equipped with an NVIDIA GeForce RTX 2080 Ti GPU. Through multiple experiments, we have identified the optimal parameter settings. When batch size = 4, lr = 0.0009, and dropout rate = 0.1, the MQN model performs the best on the IEMOCAP dataset. In the MELD dataset, the parameters should be chosen as batch size = 24, lr = 0.0008, and dropout rate = 0.1.

4.2. Results on Datasets

We trained the model on two datasets, and the final results are shown in Tables 3-5. To comprehensively analyze the results, we first examined the overall performance of the model and subsequently compared its performance for each emotional category.

4.2.1. Overall Performance

Table 3 displays the training results of all models, describing the overall performance using *accuracy* and *F1 score*. The superior results are emphasized in bold. The proposed MQN model outperforms all baseline models in terms of accuracy and F1 score on the IEMOCAP dataset. On the MELD dataset, although its accuracy is slightly lower than that of the MVN model, the F1 score of MQN still surpasses all other models by a small margin. Specifically, compared to the quantum-inspired

Table 3

Experimental results of all models on the two datasets (%).

Models	MELD		IEMOCAP	
	ACC	F1	ACC	F1
BC-LSTM	59.74	57.28	58.96	57.64
MFN	60.07	57.34	57.14	58.06
MuT	59.94	57.82	58.83	59.78
ICON	60.13	56.92	57.74	57.24
DialogueRNN	59.24	56.42	62.17	60.62
QMNN	60.23	57.83	60.98	59.18
MVN	62.15	59.83	65.32	65.44
MQN (Ours)	61.47	60.90	65.64	65.72

model QMNN, MQN achieves a 4.66% improvement in accuracy and a 6.54% improvement in F1 score on IEMOCAP. On MELD, MQN outperforms QMNN by 1.24% in accuracy and 3.07% in F1 score. Overall, the MQN model shows comparable or even better effectiveness than the existing models in conversational emotion recognition tasks. In particular, when compared to previous quantum-inspired models, MQN shows a substantial performance improvement.

We attribute the performance improvement to the following factors:

- MQN utilizes a more suitable conversation encoding method called CFE to encode dialogue features. Most methods only consider multimodal information and speaker information, neglecting utterance position information. CFE can fully integrate the useful information and establish a global interaction with multimodal features.
- The JIF method can effectively characterize the features within the modality and the interaction between modalities. The introduction of attention mechanism can fully extract the intra-modal information, and the QGRU based on the gating mechanism can better model the long-term evolution of the emotional state than the existing QRNN. In addition, the density matrix constructed by multimodal fusion also contains more information.
- Compared with classical methods, quantum theory has natural advantages in dealing with problems related to cognitive science, and conversational emotion recognition is a typical example of such problems. The process of human cognition and decision-making can show quantum-like characteristics to a certain extent, and the modeling of quantum probability provides a good explanation for this.

4.2.2. Performance for Each Class

Tables 4-5 present the performance of each emotional category obtained by all models, using *precision* and *F1 score* as evaluation metrics. Table 4 shows the training results on the IEMOCAP dataset. It is evident that our proposed MQN achieves favorable classification results for most emotion categories. For emotions such as *Happy*, *Angry*, and *Frustrated*, MQN has the highest F1 score among all models. This indicates that our model has an advantage in classifying these emotions. In addition, except for slightly lower accuracy on *Sad* and F1 score on *Excited*, MQN outperforms QMNN in all other metrics, indicating that MQN also exhibits better performance compared to the state-of-the-art quantum-inspired model. Table 5 represents the results on the MELD dataset. *Disgust* and *Fear* are missing from the table because there is an insufficient number of samples for these emotions in the MELD dataset. For the MELD dataset, the model performs well in classifying the emotions *Joy* and *Neutral*, but no significant differences are observed compared to existing models in other emotion categories. This can partly be attributed to the imbalanced distribution of emotion categories in the dataset itself. To the best of our knowledge, no quantum-inspired model has specifically addressed this issue, and related work warrants exploration in the future.

Table 4
Performance of each class on the IEMOCAP dataset (%).

Models	IEMOCAP											
	Happy		Sad		Neutral		Angry		Excited		Frustrated	
	Pre	F1	Pre	F1	Pre	F1	Pre	F1	Pre	F1	Pre	F1
BC-LSTM	32.17	37.65	71.24	61.57	58.44	54.15	58.74	63.49	68.04	64.52	58.41	61.44
MFN	45.24	36.42	70.08	68.24	50.72	54.37	61.52	67.45	61.47	63.48	53.26	60.86
MuT	47.89	41.52	74.56	65.24	55.23	51.81	63.40	58.24	68.67	66.49	56.88	62.56
ICON	34.13	30.14	71.48	68.69	53.46	56.24	68.65	61.18	65.23	59.25	59.42	59.73
DialogueRNN	82.15	29.76	79.56	73.84	59.79	58.40	61.74	57.19	66.50	69.46	55.90	62.17
QMNN	42.54	43.88	75.44	62.18	54.13	53.68	66.51	69.58	62.38	67.18	55.73	62.98
MVN	55.11	55.75	70.78	73.30	66.75	61.88	62.12	65.96	65.25	69.50	65.67	64.21
MQN (Ours)	58.89	58.56	63.86	69.93	63.43	61.58	78.49	74.68	67.87	63.52	62.50	64.84

Table 5
Performance of each class on the MELD dataset (%).

Models	MELD									
	Angry		Joy		Neutral		Sad		Surprise	
	Pre	F1	Pre	F1	Pre	F1	Pre	F1	Pre	F1
BC-LSTM	44.57	45.18	50.29	51.45	73.34	75.37	34.62	22.48	42.07	48.26
MFN	48.32	41.49	49.46	53.11	70.58	76.42	29.16	26.97	57.35	46.38
MuT	47.51	45.34	52.34	50.84	70.41	74.38	37.51	22.89	46.88	46.70
ICON	44.41	42.56	46.39	46.83	72.07	76.52	43.19	19.06	43.17	52.01
DialogueRNN	39.74	43.48	51.74	48.16	71.16	72.50	23.81	16.57	41.57	48.90
QMNN	43.68	41.02	52.97	52.85	71.88	76.91	25.80	18.16	48.14	47.38
MVN	37.96	43.55	51.54	53.41	80.65	75.10	15.77	28.62	53.46	55.58
MQN (Ours)	41.18	43.84	52.53	53.82	75.38	77.05	37.50	17.80	52.28	49.04

Table 6
Ablated MQN for both MELD and IEMOCAP datasets (%).

Models	MELD		IEMOCAP	
	ACC	F1	ACC	F1
MQN _{noEncoder}	58.16	53.39	59.52	59.40
MQN _{noQGRU}	55.12	51.27	58.12	58.98
MQN _{noQmeasure}	59.27	56.43	62.78	62.49
MQN	61.47	60.90	65.64	65.72

4.3. Ablation Study

To assess whether the components in MQN have an impact on overall performance, we conducted the ablation study for validation. By altering components of the model, several new models were proposed, namely MQN_{noEncoder}, MQN_{noQGRU}, and MQN_{noQmeasure}. MQN_{noEncoder} removes the Transformer encoder used in the MQN model for extracting intra-modal information. MQN_{noQGRU} does not include the proposed QGRU. MQN_{noQmeasure} replaces the quantum-like measurement with a fully connected layer. The final training results are shown in Table 6. As indicated in the table, the performance of MQN declines when any of these components is removed. This suggests that these components contribute to the improvement of the model's performance. Additionally, it is notable that MQN_{noQGRU} experiences the most significant performance drop, indicating that the QGRU plays a more critical role than the Transformer encoder in the JIF. This highlights the significant effectiveness of the proposed QGRU in modeling emotional evolution in conversations, as it effectively captures the interaction features between modalities.

5. Conclusions and Future Work

This work proposes MQN for conversational emotion recognition. Within a quantum-like framework, the model characterizes the representation and evolution of discourse emotion from two aspects: feature embedding and information fusion. MQN performs well on the widely

used dialogue emotion datasets MELD and IEMOCAP, outperforming the state-of-the-art quantum-inspired models. Ablation studies further indicate that the proposed JIF method and the operator-based quantum-like measurement layer are effective. To keep computation modest, MQN employs parameter sharing and a lightweight encoder, and its per-utterance mixture weights help mitigate sensitivity to missing or noisy modalities. These properties make MQN amenable to a wide range of practical application scenarios, such as emotion analysis in customer-service settings, emotion monitoring in live-streaming/short-video content, and emotion tracking in meetings and online-education platforms.

In the future, further research can be conducted on unimodal encoding by leveraging the characteristics of each modality. Additionally, we can explore more interpretable and trainable quantum-inspired fusion and measurement parameterizations, and conduct broader cross-corpus/generalization evaluations, with a particular focus on imbalanced settings.

CRedit authorship contribution statement

Zimeng Xiao: Conceptualization, Methodology, Software, Writing – review & editing; **Jia Liao:** Methodology, Software, Writing – original draft; **Jinjing Shi:** Funding acquisition, Project administration, Supervision; **Shichao Zhang:** Supervision; **Xuelong Li:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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